Name: ____________________________________

The Sky is the Limit — Decoding Weather Conditions

Student Answer Packet 1: To be completed with a partner. Numbers next to each task indicate the steps found in your ***Student Directions*** handout.

4a. Describe how the pressure on the surface of the table changed after adding **one** block to each column.

____

____

4b. The blocks are a model of columns of warm and cold air. In terms of <u>relative density</u>, explain why the height (thickness) of the warm air column is ***not*** the same as the height (thickness) of the cold air column.

____

____

6a. Compare the downward pressure exerted by the <u>two red</u> blocks (located above the card labeled **surface 1**) to the downward pressure by the <u>one blue</u> block exerted on **surface 1**.

____

____

6b. Looking at only the blocks between the table and **surface 1**, compare the downward pressure exerted by the <u>one red</u> block to the downward pressure exerted by the <u>two blue</u> blocks on the table.

____

____

6c. Compare the downward pressure exerted on the surface of the table by the three red blocks to the downward pressure exerted on the surface of the table by the three blue blocks.

____

____

7. How does the downward pressure from the red stack compare to the downward pressure from the blue stack on **surface 2**?

____

____

8a. Identify the altitude, in meters, at which the 200-mb air pressure occurred in each location.

Florida: ___________________________________

Massachusetts: __________________________________

8b. How is the rate at which air pressure changes with altitude in the atmosphere different in a column of warm air than in a column of cold air?

____

____

8c. Explain how the thickness of these two different temperature columns of air (in Florida and Massachusetts) affects the location and height of the tropopause (the top of the troposphere). The height of the tropopause occurs at approximately 370 mb.

____

____

____

9. Use the map on page 5 or 6 to plot your chosen hurricane.

10. Observe the two hurricane tracks (compass direction) plotted by you and your partner. Communicate how the track (direction) of the northern hemisphere hurricane (Andrew 8/22/92–8/27/92) was different from the track (direction) of the southern hemisphere tropical cyclone (hurricane) (Orson 4/18/89–4/23/89).

____

____

____

11. Using the *Planetary Wind Belts in the Troposphere Model* in the ***Weather Investigation Packet***, identify the name of the global wind belts that influenced the track of each of these tropical systems when these systems were classified as hurricanes.

    Northern hemisphere wind belt: ____

    Southern hemisphere wind belt: ____

12. With your partner, identify and communicate the numerical evidence, using **two** state-named locations from the *Surface Observation Map* found in this packet, that helped you determine and draw the location of both the warm front and the cold front associated with this low-pressure system. Make sure you have drawn the location of both the warm front and the cold front and have labeled the location of the center of this mid-latitude cyclone with a capital “L”.

    Warm front: ____

    ____

    Cold front: ____

    ____

13. Identify the numerical evidence from the *Thursday*, *December 15*, *2011* surface weather map in the ***Weather Investigation Packet*** that helped determine the location of the center of the low-pressure system located in Canada, north of Lake Huron.

    ____

    ____

14. The formula for temperature gradient is shown below.

$$\text{Temperature gradient} = \frac{\text{Change in temperature (°F)}}{\text{Distance (mi)}}$$

Each member of the group will calculate the temperature gradient between the boxed locations in North Dakota and Texas for **either** the *Wednesday*, *December 14*, *2011* surface weather map *or* the *Saturday*, *June 16*, *2012* surface weather map to compare the temporal change. The distance between the two locations is 1310 miles. Record your answer *to the nearest thousandth* of a °F/mi.

Then share your value with your other group member. Circle the date (*December 14, 2011* or *June 16, 2012*) of the temperature gradient that you calculated.

December 14, 2011 temperature gradient: ________________________________ °F/mi

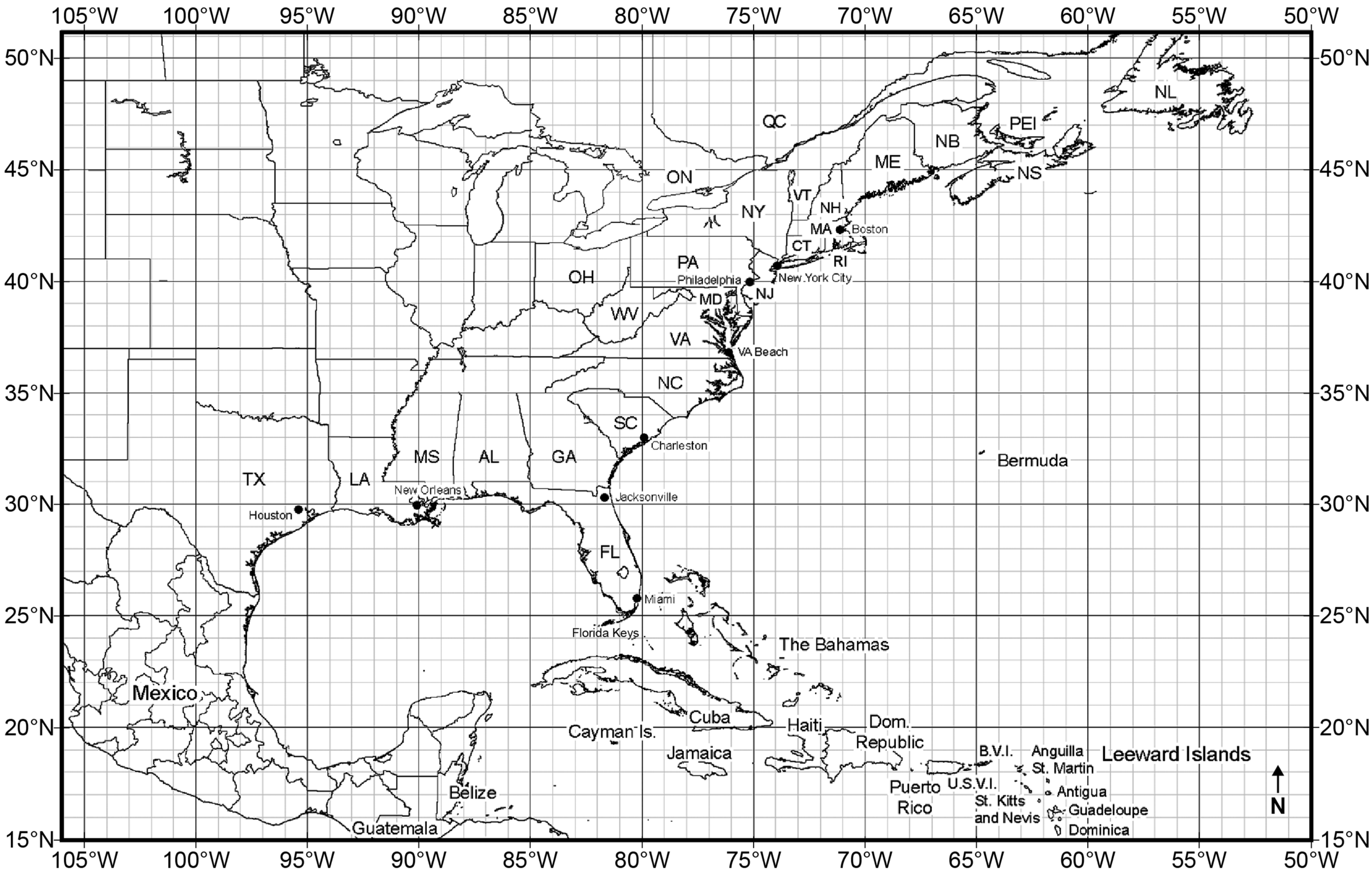
June 16, 2012 temperature gradient: _____________________________________ °F/mi

15. Using the ***Weather Investigation Packet***, collect data from the surface weather maps for Albany, NY, to complete the table below.

Date	**Temperature** (°F)	**Dewpoint** (°F)	**Wind Direction**	**Wind Speed** (knots)	**Cloud Cover** (%)	**Air Pressure** (mb)
12/14/11						
12/15/11						
12/16/11						

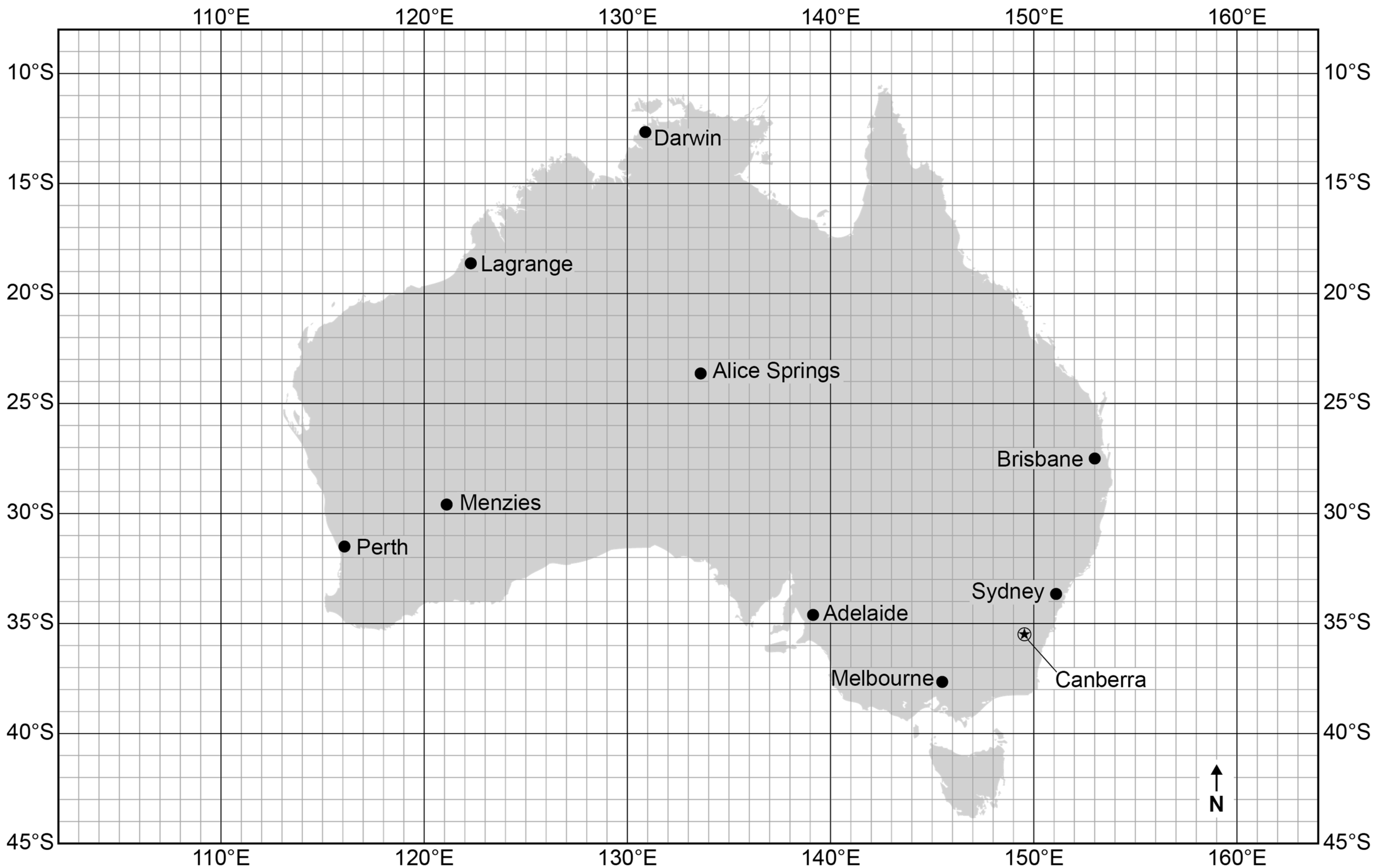
Name: ___________________________________

Hurricane Andrew Tracking Chart



Name: ___________________________________

Tropical Cyclone (Hurricane) Orson Tracking Chart



Name: ____

Surface Observation Map